

REPRODUCIBILITY IN COMPUTATIONAL EXPERIMENTS

Lucas Mello Schnorr



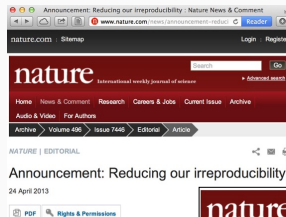
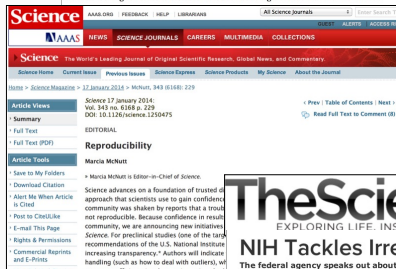
March 2026



MOTIVATION AND INTRODUCTION

PUBLIC EVIDENCE FOR A LACK OF REPRODUCIBILITY

- J.P. Ioannidis. *Why Most Published Research Findings Are False* PLoS Med. 2005.
- *Lies, Damned Lies, and Medical Science*, The Atlantic. Nov, 2010
- *Reproducibility: A tragedy of errors*, Nature, Feb 2016.
- Steen RG, *Retractions in the scientific literature: is the incidence of research fraud increasing?*, J. Med. Ethics 37, 2011



Courtesy V. Stodden,
SC, 2015

NEWSWORTHY STORIES ABOUT SCIENTIFIC MISCONDUCT

Dong-Pyou Han Assistant professor, Biomedical sciences, Iowa State University, 2013

Falsified blood results to make it appear as though a vaccine exhibited anti-HIV activity

- Han and his team received $\approx$ \$19 million from NIH
- 1 retracted publication and **resignation** of university. Sentenced in 2015 to **57 months imprisonment** for fabricating and falsifying data in HIV vaccine trials **He was also fined US \$7.2 million!**

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Dieterik Stapel Professor, Social Psychology, Univ. Amsterdam, 2011

I failed as a scientist. I adapted research data and fabricated research. Not once, but several times, not for a short period, but over a longer period of time. [...] I am aware of the suffering and sorrow that I caused to my colleagues... I did not withstand the pressure to score, to publish, the pressure to get better in time. I wanted too much, too fast. In a system where there are few checks and balances, where people work alone, I took the wrong turn.

58 retracted publications

NEWSWORTHY STORIES ABOUT SCIENTIFIC MISCONDUCT

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58 retracted publications

Brian Wansink Professor, Psychological Nutrition, Cornell, 2016

When she arrived, I gave her a data set of a self-funded, failed study which had null results. I said "This cost us a lot of time and our own money to collect. There's got to be something here we can salvage because it's a cool (rich & unique) data set." I told her what the analyses should be and what the tables should look like. [...] Every day she came back with puzzling new results, and every day we would scratch our heads, ask "Why," and come up with another way to reanalyze the data with yet another set of plausible hypotheses

17 retracted publications

A CREDIBILITY CRISIS?

Scientific misconduct is obviously wrong but it's **not new!**

- Every domain has its black sheep
- The publish or perish pressure is a huge pain

Media attention **inflates conspiracy opinions** 😞

Scientific result are worthless. Stop the scientific dictatorship/lobby!

The Battle against Scientific Fraud

CNRS International Magazine



A CREDIBILITY CRISIS?

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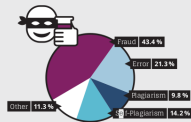
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Biomedical fraud in figures

Cause of retraction 1977 to 2012



Number of publications and retractions 1977 to 2013



Fraud is the (**uninteresting**) visible part of the iceberg

- **Failing to reproduce** the results of others is common

1,500 scientists lift the lid on reproducibility, Nature, May 2016

- How so? **Why now?** **Why is this important?** What can we do about it?

The Battle against Scientific Fraud

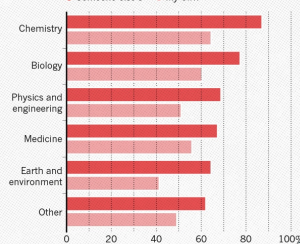
CNRS International Magazine



HAVE YOU FAILED TO REPRODUCE AN EXPERIMENT?

Most scientists have experienced failure to reproduce results.

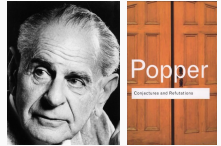
• Someone else's • My own



REPRODUCIBILITY OF EXPERIMENTAL RESULTS IS THE *HALLMARK OF SCIENCE*

1934: Karl Popper puts the notions of falsifiability and crucial experiment as the hallmark of science

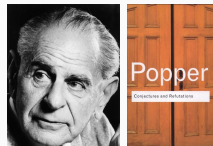
- If no experiment can be set up to disprove your theory, it is not science
- Good experiments discriminate good theories from bad ones
- Non-reproducible single occurrences are of no significance to science



REPRODUCIBILITY OF EXPERIMENTAL RESULTS IS THE *HALLMARK OF SCIENCE*

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- Good experiments **discriminate good theories from bad ones**
- **Non-reproducible** single occurrences are of no significance to science



An ideal rather than the norm

Popper's proposal works well for Physics from the 18th century but is not so simple for many other domains:

- Theory of evolution
- Spotting a SuperNova
- Particle Physics (a single LHC)
- Biology (every animal does not behave in the same way)
- Anthropology (impact on people from a remote culture)

REPRODUCIBILITY: A CORE VALUE OF SCIENCE

1. Universality: Science aims for objective findings, accessible to anyone
Reproducibility acts as a Universality/Robustness control
2. Incremental: We build on each others work but everybody makes mistakes
Methods, biases, ... How to discriminate sound theories experiments from bad ones? 😊
Reproducibility acts as a Quality control

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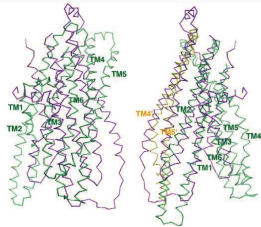
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Reproducibility acts as a Quality control

But, scientific practices have greatly evolved, in particular since we rely on computers

How computers broke science – and what we can do about it
– Ben Marwick, The conversation, 2015



HOW COMPUTERS BROKE SCIENCE



Geoffrey Chang (Scripps, UCSD) works on crystallography and studies the structure of cell membrane proteins.

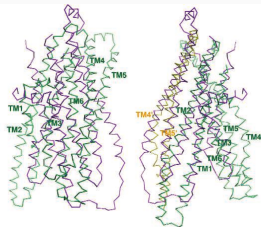
He specialized in structures of **multidrug resistant transporter proteins in bacteria**: MsbA de Escheria Choli (Science, 2001), Vibrio cholera (Mol. Biology, 2003), Salmonella typhimurium (Science, 2005)

2006: Inconsistencies reveal **a programming mistake**

A homemade data-analysis program had flipped two columns of data, inverting the electron-density map from which his team had derived the protein structure.

5 retracts that motivate **improved software engineering practices** in comp. biology

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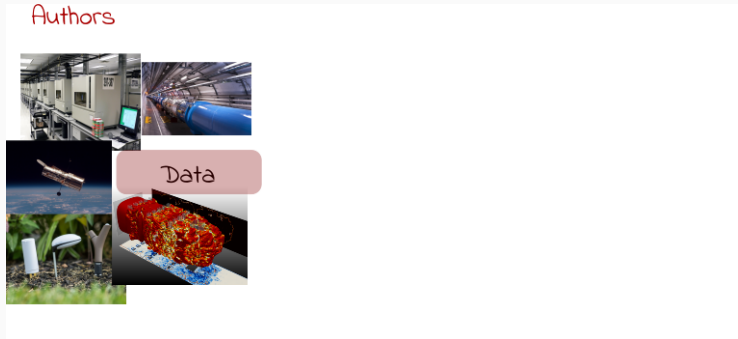
5 retracts that motivate **improved software engineering practices** in comp. biology

There is **worse!**

- The generalized and intensive use of **spreadsheets** (**COVID tracing**)
- Relying on **black box** statistical methods is infinitely easier than understanding them
 - Learning and Data Analytics frameworks are nuclear weapons
- **Numerical errors** and **software environment** unawareness

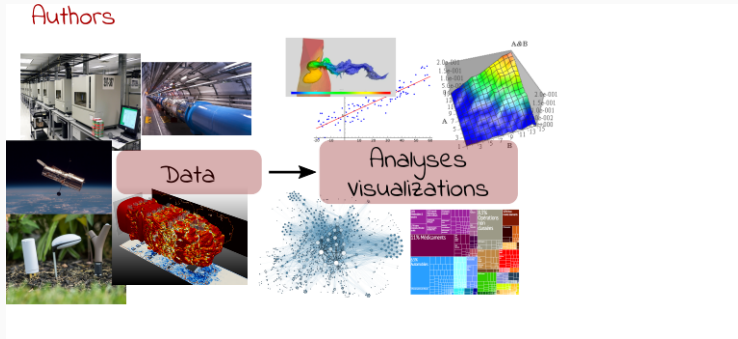
MODERN SCIENCE

The processing steps between raw observations and findings have gotten increasingly numerous and complex.



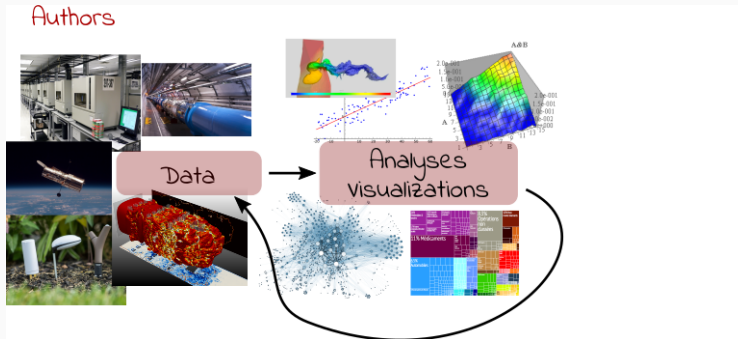
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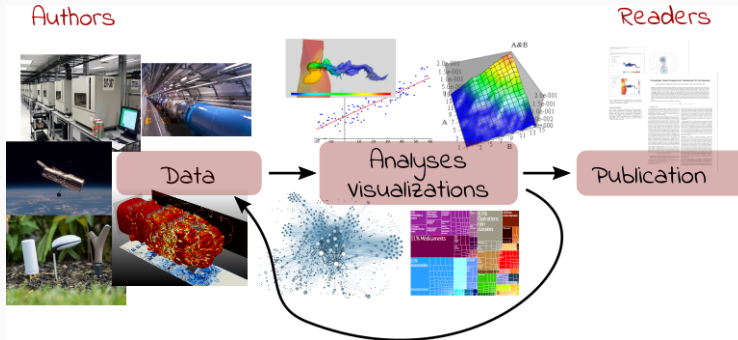
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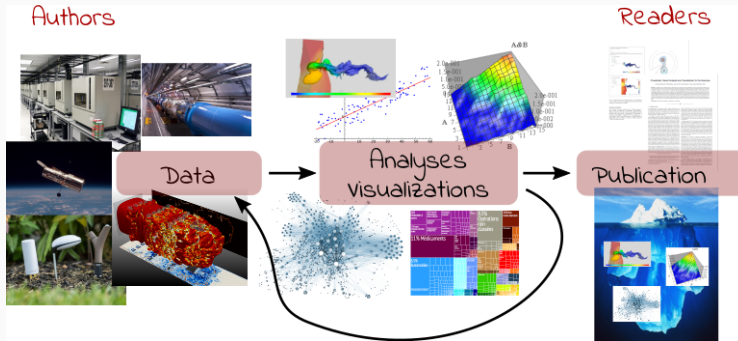
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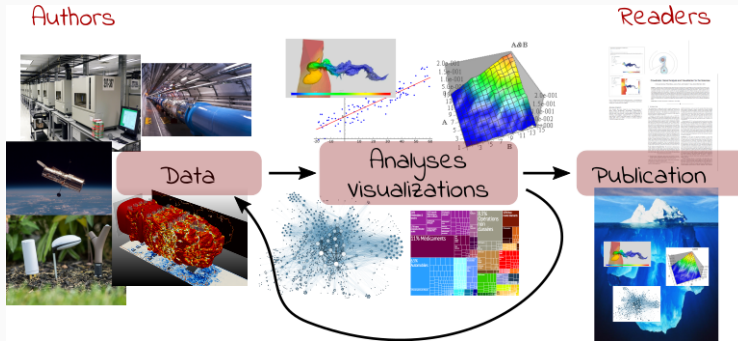
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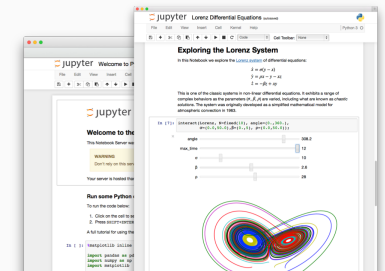


Reproducible Research = Bridging the Gap by working Transparently

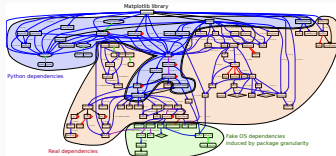
TOOLS

EXISTING TOOLS, EMERGING STANDARDS

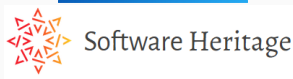
Notebooks and workflows



Software environments



Sharing platforms



Un document computationnel

Mon ordinateur m'indique que π vaut *approximativement*

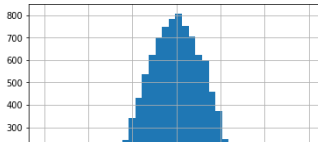
3.141592653589793

Mais calculé avec la **méthode** des [aiguilles de Buffon](#), on obtiendrait comme **approximation** :

```
import numpy as np
N = 1000000
x = np.random.uniform(size=N, low=0, high=1)
theta = np.random.uniform(size=N, low=0, high=np.pi/2)
2/(sum((x+np.sin(theta))>1)/N)
```

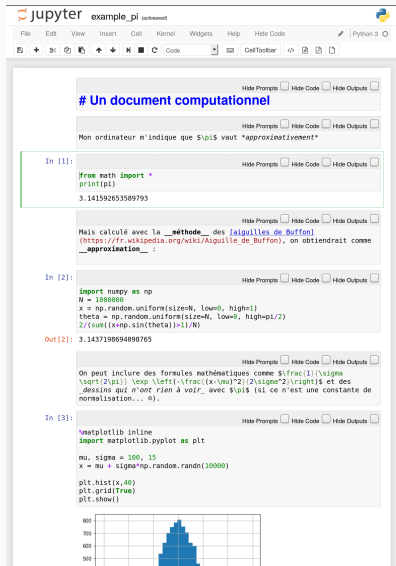
3.1437198694098765

On peut inclure des formules mathématiques comme $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ et des *dessins* qui n'ont rien à voir avec π (si ce n'est une constante de normalisation... ☺).



TOOL 1: COMPUTATIONAL NOTEBOOKS/LITERATE PROGRAMMING

Document initial dans son environnement



The screenshot shows a Jupyter Notebook interface with the title 'example_pi'. The notebook contains several cells:

- A text cell with the heading '# Un document computationnel'.
- A text cell with the sentence 'Mon ordinateur m'indique que π vaut *approximativement*'.
- A code cell (In [1]:) that prints the value of π using `print(pi)`, resulting in `3.141592653589793`.
- A text cell with a paragraph about the Buffon's needle method, including a link to Wikipedia.
- A code cell (In [2]:) that uses NumPy to simulate the Buffon's needle method, resulting in the value `3.1437198694098765`.
- A text cell with a paragraph about mathematical formulas and plots.
- A code cell (In [3]:) that uses Matplotlib to create a histogram of random data, showing a bell-shaped distribution.

Document final

Un document computationnel

Mon ordinateur m'indique que π vaut *approximativement*

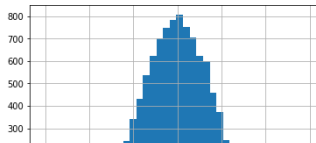
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```
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2/(sum((x+np.sin(theta))>1)/N)
```

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On peut inclure des formules mathématiques comme $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ et des *dessins* qui n'ont rien à voir avec π (si ce n'est une constante de normalisation... ☺).



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The screenshot shows a Jupyter Notebook titled 'example_pi'. The interface includes a top menu bar with 'File', 'Edit', 'View', 'Insert', 'Cell', 'Kernel', 'Widgets', 'Help', 'Hide Code', and 'Python 3'. Below the menu is a toolbar with icons for saving, undo, redo, and other actions. The notebook content is organized into cells. The first cell is a text cell with the title '# Un document computationnel' and a paragraph: 'Mon ordinateur m'indique que π vaut approximativement 3.141592653589793'. The second cell is a code cell with the following code:

```
from math import *
print(pi)
```

 The output of this cell is '3.141592653589793'. The third cell is a text cell with the paragraph: 'Mais calculé avec la méthode des aiguilles de Buffon (https://fr.wikipedia.org/wiki/Aiguille_de_Buffon), on obtiendrait comme approximation :'. The fourth cell is a code cell with the following code:

```
import numpy as np
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x = np.random.uniform(size=N, low=0, high=1)
theta = np.random.uniform(size=N, low=0, high=pi/2)
2/(sum((x*np.sin(theta))>1)/N)
```

 The output of this cell is '3.1437198694098765'. The fifth cell is a text cell with the paragraph: 'On peut inclure des formules mathématiques comme $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ et des dessins qui n'ont rien à voir avec π (si ce n'est une constante de normalisation... 0)'. The sixth cell is a code cell with the following code:

```
%matplotlib inline
import matplotlib.pyplot as plt

mu, sigma = 100, 15
x = mu + sigma*np.random.randn(10000)

plt.hist(x,40)
plt.grid(True)
plt.show()
```

 The output of this cell is a histogram plot showing a normal distribution curve.

Mark Down

Document final

Un document computationnel

Mon ordinateur m'indique que π vaut *approximativement*

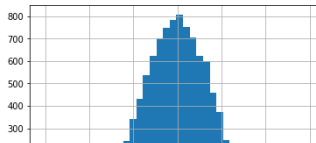
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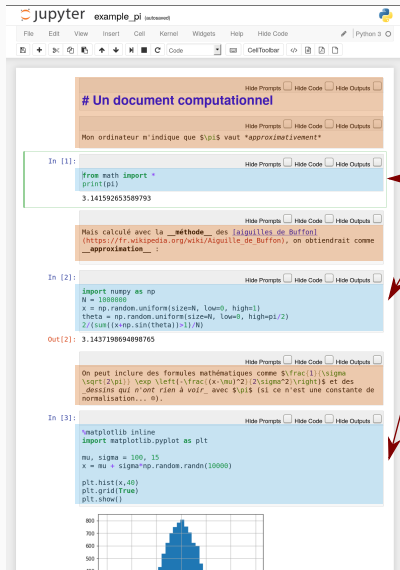
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On peut inclure des formules mathématiques comme $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ et des *dessins* qui n'ont rien à voir avec π (si ce n'est une constante de normalisation... 0).



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The screenshot shows a Jupyter Notebook interface with the title 'example_pi'. The notebook contains several cells:

- A text cell with the title '# Un document computationnel' and a paragraph: 'Mon ordinateur m'indique que π vaut approximativement 3.141592653589793'.
- A code cell (In [1]) that prints the value of π :

```
from math import *\nprint(pi)
```
- A text cell explaining the Buffon's needle method: 'Mais calculé avec la méthode des aiguilles de Buffon (https://fr.wikipedia.org/wiki/Aiguille_de_Buffon), on obtiendrait comme approximation :'. A red arrow labeled 'Code' points to the code cell below.
- A code cell (In [2]) that implements the Buffon's needle method using NumPy:

```
import numpy as np\nN = 1000000\nx = np.random.uniform(size=N, low=0, high=1)\ntheta = np.random.uniform(size=N, low=0, high=pi/2)\n2/(sum((x*np.sin(theta))>1)/N)
```
- An output cell (Out [2]) showing the result: 3.1437198694098765.
- A text cell explaining the inclusion of mathematical formulas and plots: 'On peut inclure des formules mathématiques comme $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ et des dessins qui n'ont rien à voir avec π (si ce n'est une constante de normalisation... $\odot$)'.
- A code cell (In [3]) that uses Matplotlib to create a histogram:

```
%matplotlib inline\nimport matplotlib.pyplot as plt\nmu, sigma = 100, 15\nx = mu + sigma*np.random.randn(10000)\nplt.hist(x,40)\nplt.grid(True)\nplt.show()
```
- An output cell showing a histogram of the generated data, which is a bell-shaped curve centered around 100.

Document final

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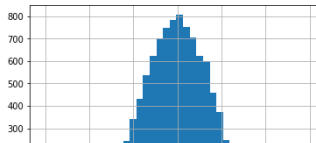
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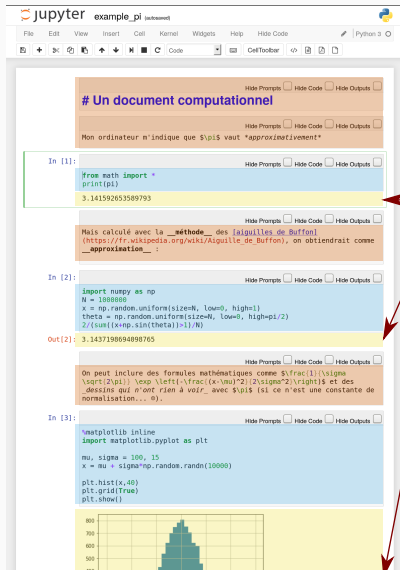
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The output is 3.141592653589793.
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- A code cell (In [2]) that implements the Buffon's needle method using NumPy:

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2/(sum((x*np.sin(theta))>1)/N)
```

The output (Out [2]) is 3.1437198694098765.
- A text cell explaining that mathematical formulas can be included using LaTeX syntax, such as $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
- A code cell (In [3]) that uses Matplotlib to plot a histogram of random data:

```
%matplotlib inline
import matplotlib.pyplot as plt

mu, sigma = 100, 15
x = mu + sigma*np.random.randn(10000)

plt.hist(x,40)
plt.grid(True)
plt.show()
```

The output is a histogram showing a normal distribution centered around 100.

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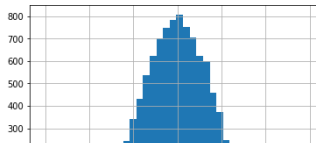
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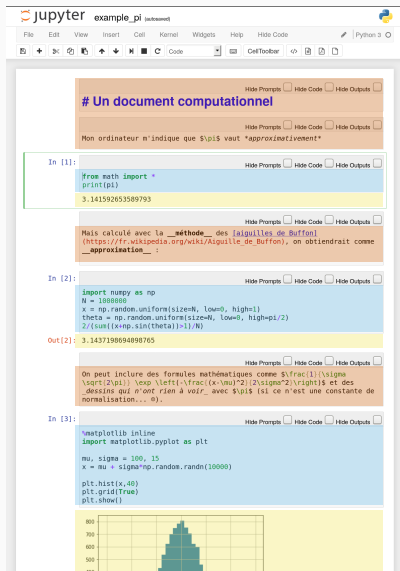
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```

 The output (Out[2]:) is 3.1437198694098765.
- A text cell explaining that mathematical formulas and plots can be included.
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```
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x = mu + sigma*np.random.randn(10000)  
  
plt.hist(x,40)  
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plt.show()
```

 The output is a histogram of the generated data.

Document final

Un document computationnel

Mon ordinateur m'indique que π vaut *approximativement*

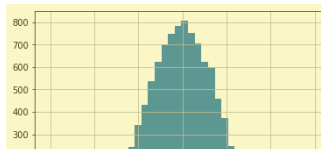
3.141592653589793

Mais calculé avec la **méthode** des [aiguilles de Buffon](https://fr.wikipedia.org/wiki/Aiguille_de_Buffon), on obtiendrait comme **approximation** :

```
import numpy as np  
N = 1000000  
x = np.random.uniform(size=N, low=0, high=1)  
theta = np.random.uniform(size=N, low=0, high=pi/2)  
2/(sum((x*np.sin(theta))>1)/N)
```

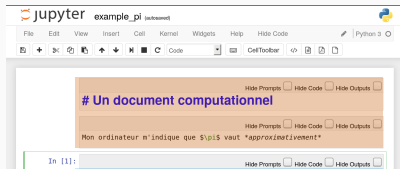
3.1437198694098765

On peut inclure des formules mathématiques comme $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ et des *dessins* qui n'ont rien à voir avec π (si ce n'est une constante de normalisation... ☺).



TOOL 1: COMPUTATIONAL NOTEBOOKS/LITERATE PROGRAMMING

Document initial dans son environnement



Document final

Un document computationnel

Mon ordinateur m'indique que π vaut *approximativement*

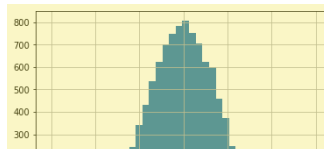
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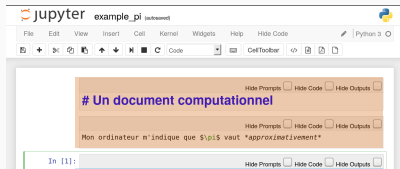


Export



TOOL 1: COMPUTATIONAL NOTEBOOKS/LITERATE PROGRAMMING

Document initial dans son environnement



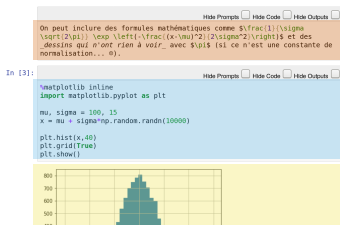
Document final

Un document computationnel

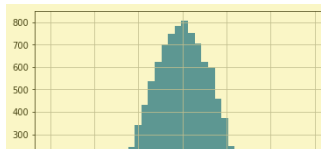
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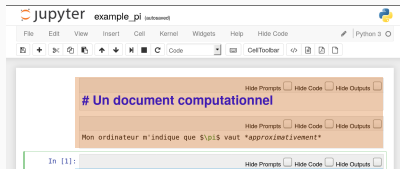


des $\sigma\sqrt{2\pi}$ et $\frac{1}{\sigma\sqrt{2\pi}}$ des dessins qui n'ont rien à voir avec π (si ce n'est une constante de normalisation... ☺).



TOOL 1: COMPUTATIONAL NOTEBOOKS/LITERATE PROGRAMMING

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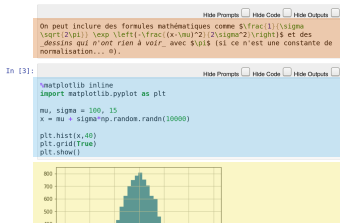
Document final

Un document computationnel

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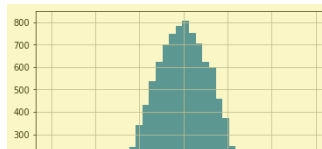
3.141592653589793

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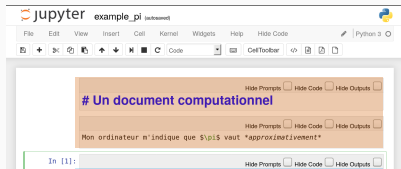
$\sigma\sqrt{2\pi}$ * $\frac{1}{\sqrt{2\sigma^2}}$

des *dessins* qui n'ont rien à voir avec π (si ce n'est une constante de normalisation... ☹).



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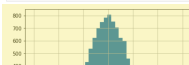


On peut inclure des formules mathématiques comme $\sqrt{2\pi}$ et des dessins qui n'ont rien à voir avec π (si ce n'est une constante de normalisation... ☺).

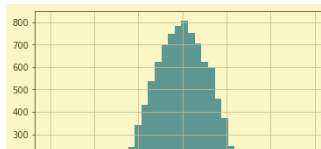
```
In [3]: %matplotlib inline
import matplotlib.pyplot as plt

mu, sigma = 100, 15
x = mu + sigma*np.random.randn(10000)

plt.hist(x,40)
plt.grid(True)
plt.show()
```



des $\sigma\sqrt{2\pi}$ $\frac{1}{\sigma\sqrt{2\pi}}$ dessins qui n'ont rien à voir avec π (si ce n'est une constante de normalisation... ☺).



TOOL 2: FIGHTING SOFTWARE ENVIRONMENTS NIGHTMARE

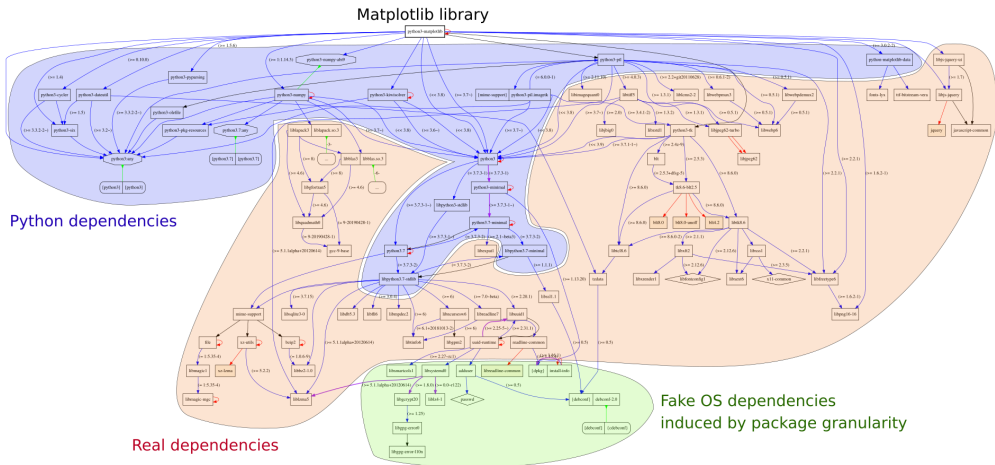
What is hiding behind a simple

```
1 Package: python3-matplotlib
2 Version: 2.1.1-2
3 Depends: python3-dateutil, python-matplotlib-data (>= 2.1.1-2),
4 python3-pyparsing (>= 1.5.6), python3-six (>= 1.10), python3-tz,
5 libjs-jquery, libjs-jquery-ui, python3-numpy (>= 1:1.13.1),
6 python3-numpy-abi9, python3 (<< 3.7), python3 (>= 3.6~),
7 python3-cycler (>= 0.10.0), python3:any (>= 3.3.2-2~), libc6 (>=
8 2.14), libfreetype6 (>= 2.2.1), libgcc1 (>= 1:3.0), libpng16-16 (>=
9 1.6.2-1), libstdc++6 (>= 5.2), zlib1g (>= 1:1.1.4)
```

TOOL 2: FIGHTING SOFTWARE ENVIRONMENTS NIGHTMARE

What is hiding behind a simple

```
1 package: python2-matplotlib
```



TOOL 2: FIGHTING SOFTWARE ENVIRONMENTS NIGHTMARE

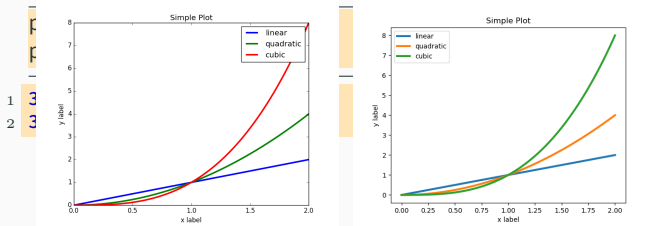
Python and its rapidly evolving environment

```
python2 -c "print(10/3)"  
python3 -c "print(10/3)"
```

```
1 3  
2 3.3333333333333335
```

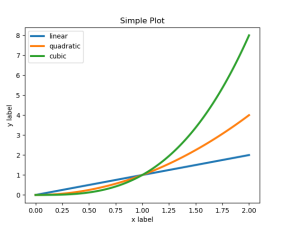
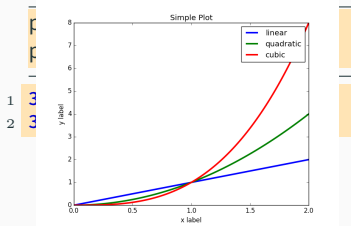
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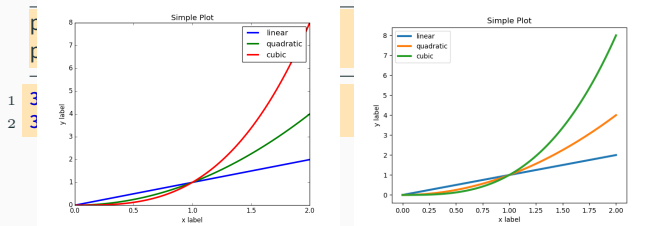
Python and its rapidly evolving environment



Cortical Thickness Measurements (PLOS ONE, June 2012): *FreeSurfer: differences were found between the Mac and HP workstations and between Mac OSX 10.5 and OSX 10.6.*

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Python and its rapidly evolving environment



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TOOL 3: FIGHTING INFORMATION LOSS WITH ARCHIVES

D. Spinellis. [The Decay and Failures of URL References](#). CACM, 46(1), Jan 2003.

The half-life of a referenced URL is approximately 4 years from its publication date.

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Applied Clinical Informatics. 4 (4), 2013

half life ranged from 2.2 years in EMHJ to 5.3 years in BMJ

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Article archives  

Data archives  

Software Archive  Software Heritage



or



= awesome collaborations $\neq$ archive

REPRODUCIBILITY

DIFFERENT (BUT CONVERGING) REPRODUCIBILITY CONCERNS

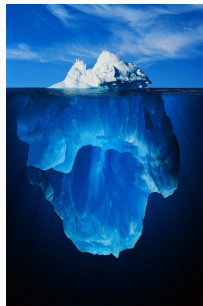
Reproducibility/robustness of the scientific fact, the statistical analysis, the computation, the observation, the process, ... ?

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Bad computer/statistic/publication practices "broke science" 😊

- Ensure articles and data are **available**
 - **Open Access**, **Open Data**, FAIR and DMP

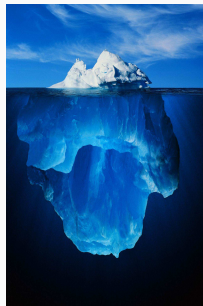


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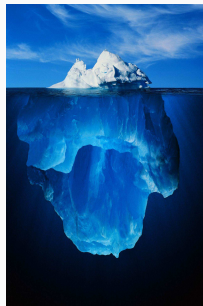


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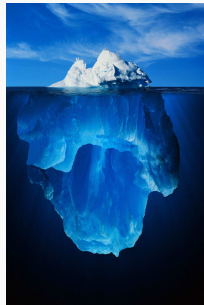


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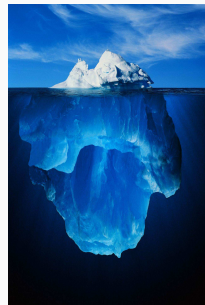


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This requires **first class software engineering practices** instead of building on prototypes

Software factories, Archives, and Provenance Tracking tools

CLOSURE

CHANGING RESEARCH PRACTICES

Soft. Engineering, Statistics, and Reproducible Research in the **curricula**



- **Book on RR** *Vers une recherche reproductible: Faire évoluer ses pratiques*
- **MOOC on RR** (3rd edition Feb. 2020)
- **A new "Advanced RR" MOOC** (1st edition 2024)
 - Managing research data (HDF5, archiving)
 - Software environment control (Docker, Guix)
 - Scientific workflow (snakemake)

Manifesto: "*I solemnly pledge*" (**WSSSPE**, **Lorena Barba**, **FAIR**)

1. I will teach my graduate students about reproducibility
2. All our research code (and writing) is under version control
3. We will always carry out verification and validation
4. We will share data, plotting script & figure under CC-BY
5. We will upload the preprint to arXiv at the time of submission of a paper
6. We will release code at the time of submission of a paper
7. We will add a "Reproducibility" declaration at the end of each paper
8. I will keep an up-to-date web presence

Artifact evaluation and ACM badges



Major conferences

- **Supercomputing**: Artifact Description (AD) **mandatory**, Artifact Evaluation (AE) still **optional**, **Double blind** vs. **RR**
- **NeurIPS**, **ICLR**: **open reviews**, reproducibility challenge



Joelle Pineau @ NeurIPS'18

- **ACM SIGMOD 2015-2019**, Most Reproducible Paper Award...

Mentalities are evolving people care, make stuff available, errors are found and fixed

RESOURCES AND ACKNOWLEDG- MENTS

RESOURCES AND ACKNOWLEDGMENTS

Slides are available here

https://github.com/schnorr/SMPE/raw/master/lectures/talk_26_03_20_SemIntegracaoPPGC.pdf

Thanks to

- Arnaud Legrand for sharing slides and knowledge about this topic
- My laboratory students that have understood the importance
 - It may be hard to face/learn new things, but they are essential to evolve!

MOOC Reproducible Research II: Practices and tools for managing computations and data,
Learning Lab Inria

- Konrad Hinsén, Arnaud Legrand, Christophe Pouzat
- 3rd Edition: May 2026 – December 2026

